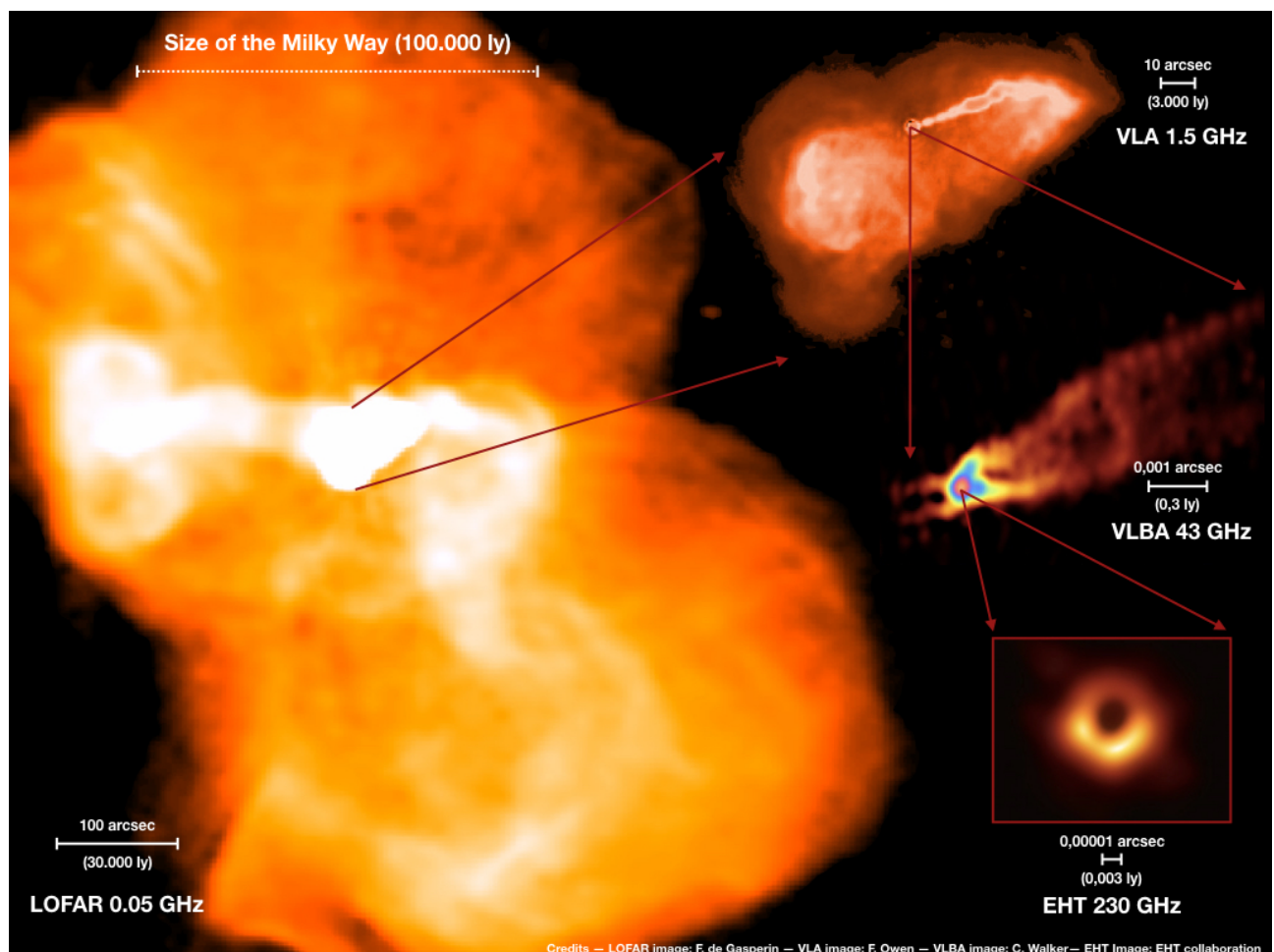


# HS1 Radio Astronomy – Background

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The radio galaxy M 87 (Virgo A) observed with the LOw Frequency ARray (LOFAR)



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# 1 Basics of radio astronomy

## 1.1 Radio waves

The radio wave range extends from about  $\lambda = 1$  mm to  $>1$  km, and is subdivided in mm, cm and m bands. Radio waves are usually specified in frequencies  $\nu$ , where  $c = \lambda \nu$ , and  $c$  is the speed of light. The approximate frequency range of the *radio window* extends from about 15 MHz ( $\lambda 20$  m) to 600 GHz ( $\lambda 0.5$  mm).

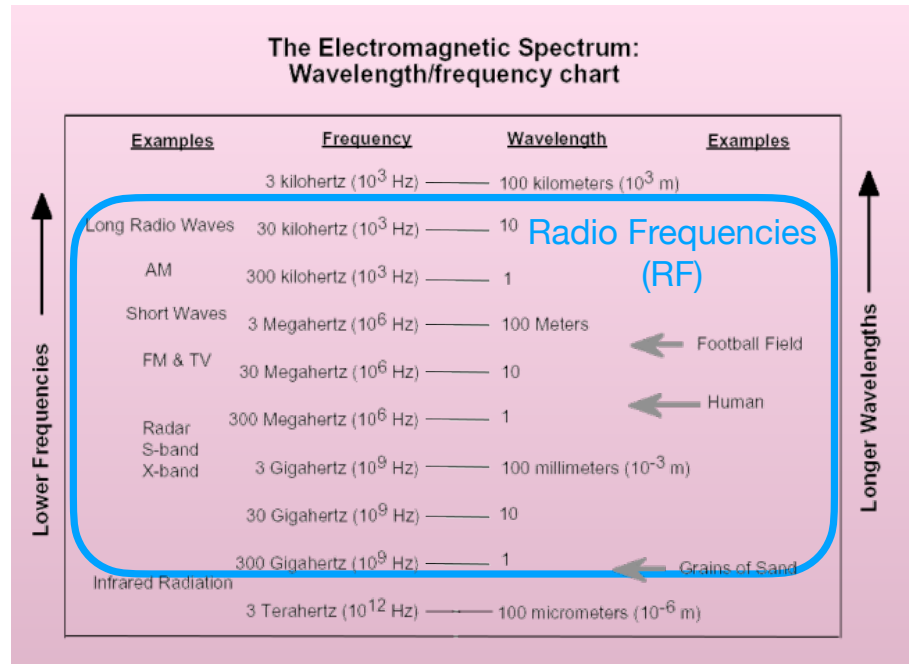


Figure 1: Wavelengths and frequencies of the radio window.

## 1.2 Observation window

Radio waves can be received on Earth in the wavelength range between a few centimetres and several metres (Fig. 1). The transmission of the atmosphere amounts to  $>90\%$  between  $\lambda 5$  cm and  $\lambda 15$  m (20 MHz–6 GHz). In the sub-millimetre and millimetre band, as well as in the short wavelength centimetre band, the opacity of the Earth's atmosphere increases strongly due to molecular bands of water vapour and oxygen. Observations are only possible among some windows from high altitude locations (Fig. 2).

On the other hand, at metre wavelengths, the transmission of radio waves is limited by the ionosphere. The ionosphere is a layer of the Earth's atmosphere at a height of about 100 km. Consider for comparison that the troposphere (the 'weather layer') extends only up to 12 km. In the ionosphere, a high degree of ionisation prevails, so that radio waves are scattered and absorbed by free electrons.

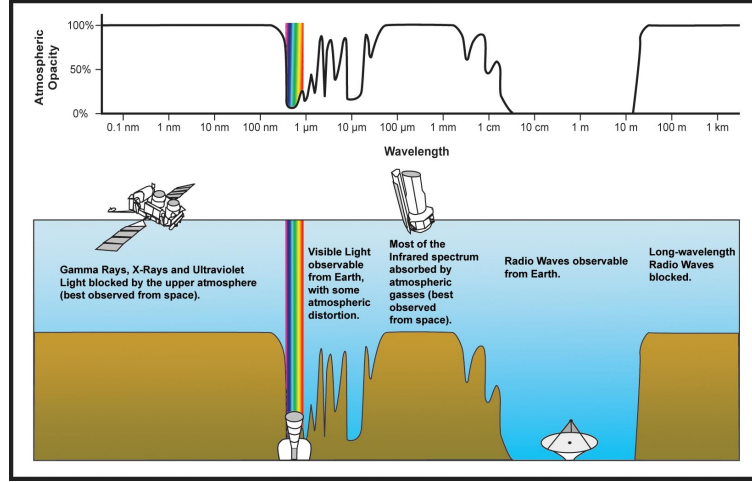


Figure 2: Transmission of the Earth's atmosphere for electromagnetic radiation.

### 1.3 Spatial resolution of single-dish telescopes and radio interferometers

Single-dish antennae have a relatively poor spatial resolution, which is given by the diffraction limit of the telescope. This limit is valid for all telescopes, e.g. also for optical ones, and is proportional to the wavelength. Thus, it is a limitation especially for radio astronomy.

A circular aperture diffracts the incident radiation. The diffraction pattern consists of a bright central disc and fainter outer rings. The angular diameter of the central disc, defined by the points where the intensity becomes zero, is given by  $q = 2.44\lambda / D$ , where  $D$  is the diameter of the aperture,  $\lambda$  the wavelength, and  $q$  the angular diameter in radians<sup>1</sup>.

The diffraction limit for a circular aperture describes the minimum angular distance that two point sources must have in order to be perceived as separate objects. The so-called Rayleigh criterion requires that the centre of the central disc of the first object must be farther away than the edge of the central disc of the second object. In radio astronomy, the half-width of the central disc (main lobe) is often adopted as the resolution. The half-width is the width of the disc at half of the intensity maximum (full width at half-maximum, or FWHM).

**Thus, the theoretical spatial resolution of a radio telescope is calculated by:**

$$q = 1.02 \frac{\lambda}{D} \text{ rad}, \quad (1)$$

**where  $q$  is the FWHM of the main lobe in units of radians.** Expressed in degrees, this gives:

$$q = 58.4^\circ \frac{\lambda}{D}. \quad (2)$$

*Exercise.* Calculate the resolving power of a 10 cm optical telescope at  $\lambda 550 \text{ nm}$  and of the 100-m Effelsberg telescope at  $\lambda 1.3 \text{ cm}$ .

<sup>1</sup>  $1 \text{ rad} = 206265'' = 57.3^\circ$

Compare this with the apparent size of Sun and Moon (each 30 arcminutes) and Venus (20 arcseconds), the resolving power of an unaided eye (1 arcminute) and the size of the astronomical seeing in the visible light (2 arcseconds).

For a radio interferometer, the spatial resolution is given by the largest distance of its individual antennae from each other. These distances are called baselines and thus with the baseline length  $B$  we obtain:

$$q \approx \lambda B^{-1} \text{ rad}, \quad (3)$$

or, for practical applications, in useful units:

$$q \approx 2063 \lambda B^{-1} \text{ mas}, \quad (4)$$

with  $\lambda$  in cm,  $B$  in km, and  $q$  in mas (milli-arcseconds). The Event Horizon Telescope (EHT) observed the supermassive black hole in the galaxy M87 at  $\lambda$  1.3 mm with a maximum baseline length of 10,000 km.

*Exercise.* Calculate the theoretical resolution of the EHT.

## 2 Radiation theory

### 2.1 Power and intensity

The radio radiation is measured as power at the receiver. The basic geometry is illustrated in Fig. 3.

#### 2.1.1 General case

The **infinitesimal power**  $dP$ , which is applied to the area  $dA$  with a solid angle element  $d\Omega$  is:

$$dP = I_\nu \cos \Theta d\Omega dA d\nu \quad (5)$$

$dP$  = power (Watt)

$I_\nu$  = electromagnetic radiation intensity ( $\text{W m}^{-2} \text{Hz}^{-1} \text{sr}^{-1}$ )

$d\nu$  = infinitesimal bandwidth

$\Theta$  = angle between  $d\Omega$  and zenith

while the solid angle  $d\Omega$  is measured in steradians (sr).  $I_\nu$  is called **intensity or brightness**. As a synonym to  $I_\nu$  also  $B_\nu$  is used, especially if the intensity is modelled with the Planck-function. The frequency intervall lies between  $\nu$  and  $\nu + d\nu$ . It holds:

$$dP = A I_\nu \cos \Theta d\Omega d\nu, \quad (6)$$

if  $I_\nu$  does not depend on  $A$ .

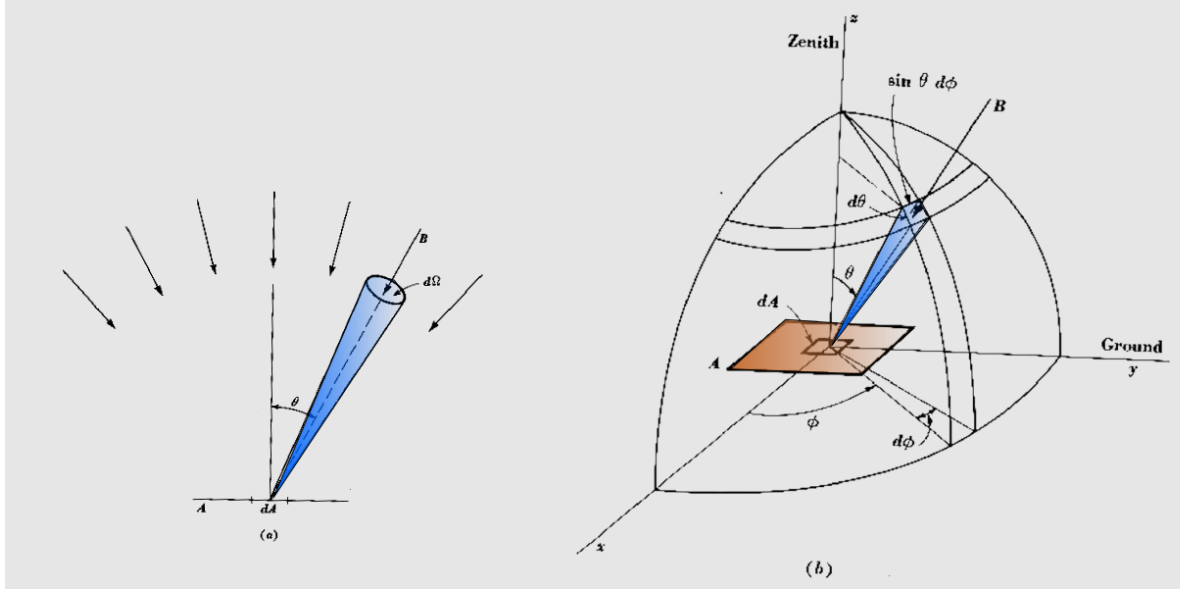


Figure 3: Definition of basic spherical geometry. *Left:* The line of sight (LOS)  $\vec{B}$  within the infinitesimal solid angle element  $d\Omega$ . *Right:* Position of  $\vec{B}$  and the celestial sphere with spherical coordinates. *Credit:* Kraus (1986).

The **total power**  $P$  is obtained integrating:

$$P = A \int_v^{v+\Delta v} \iint_{\Omega} I_v \cos \Theta \, d\Omega \, dv. \quad (7)$$

Note that the intensity  $I_v$  is a function of the direction and frequency. The **radio continuum spectrum** is the variation of  $I_v$  with frequency  $v$ .

Integrated over the bandwidth  $\Delta v$ :

$$I = \int_v^{v+\Delta v} I_v \, dv, \quad (8)$$

where  $I$  is the **integrated intensity** in  $\text{W m}^{-2} \text{sr}^{-1}$  and  $\Delta v$  is the bandwidth in Hz. If one integrates over the entire radio range, one obtains the **total intensity** or **total radio brightness**.

Analogous to equations (5)–(7) the **spectral power**  $P_v = dP/dv$  becomes:

$$dP_v = I_v \cos \Theta \, d\Omega \, dA \quad (9)$$

$$= A I_v \cos \Theta \, d\Omega, \quad (10)$$

and thus, integrating over the solid angle  $d\Omega$ :

$$P_v = A \iint_{\Omega} I_v \cos \Theta \, d\Omega. \quad (11)$$



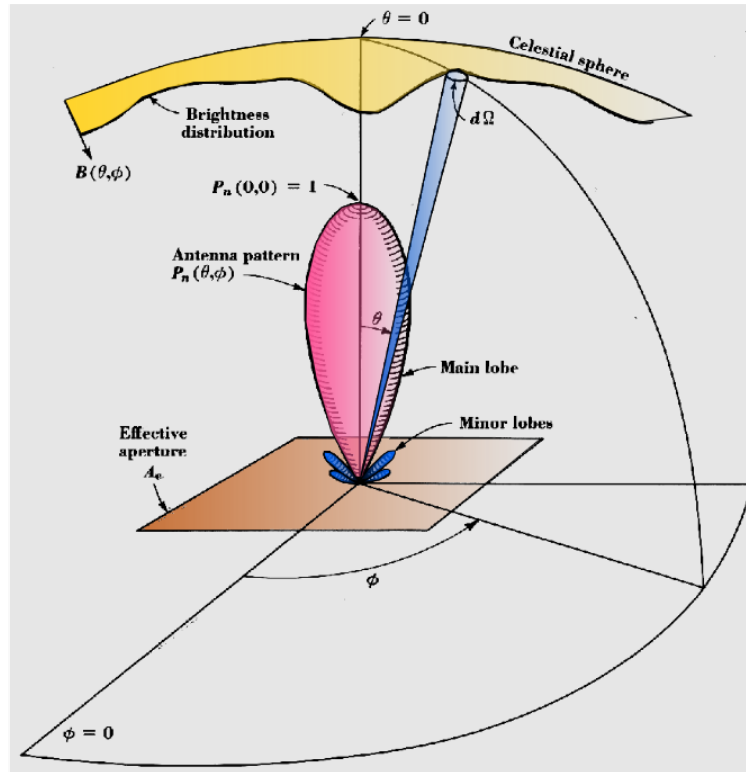


Figure 4: Normalised distribution of the directional sensitivity of the antenna *main lobe* and *side* (or *minor lobes*). *Credit: Kraus (1986)*.

### 2.1.2 Power from a hemisphere at constant intensity

Given:

$$d\Omega = \sin \Theta \, d\varphi \, d\Theta, \quad (12)$$

and  $I_V = I_c$  constant. Substitution into equation (11) and integrating over the hemisphere, gives:

$$P_V = A I_c \int_0^{2\pi} \int_0^{\pi/2} \cos \Theta \sin \Theta d\varphi d\Theta. \quad (13)$$

That results in:

$$P_V = \pi A I_c, \quad (14)$$

in units of  $\text{W Hz}^{-1}$ . This is the received spectral power from an entire hemisphere with solid angle  $\Omega = 2\pi$ . Note that, even if the solid angle of a hemisphere is  $2\pi$ , the integral in equation (13) gives only a factor  $\pi$ .

Multiplication with the bandwidth finally results in the **total power**:

$$P = \pi A I_c \Delta v. \quad (15)$$

## 2.2 When using a radio antenna

Consider a source with an intensity distribution  $I_V = I_V(\Theta, \varphi)$ , where the antenna has a directional sensitivity that can be described by a normalised power pattern  $P_n(\Theta, \varphi)$ . The spectral power is given by equation (11):

$$P_V = A \iint_{\Omega} I_V(\Theta, \varphi) \cos \Theta d\Omega. \quad (16)$$

If we take into consideration the following changes:

- The area  $A$  is replaced by the **effective antenna area**  $A_e$ , which is smaller than the geometrical area of the antenna. Due to the reception losses of a real antenna, the received power does not correspond to the geometrical area  $A$  but to a smaller area  $A_e$  of an idealized lossless antenna;
- The directional sensitivity of the antenna is described by the distribution function  $P_n(\Theta, \varphi)$ , which is dimensionless and normalised, thus:

$$P_n(0, 0) = 1. \quad (17)$$

The normalised power pattern  $P_n(\Theta, \varphi)$  replaces the factor  $\cos \Theta$ ;

- An antenna can only receive one polarisation direction. Therefore a factor 1/2 must be applied because of polarisation;

the appropriately modified and integrated equation (16) becomes:

$$P_V = \frac{1}{2} A_e \iint_{\Omega} I_V(\Theta, \varphi) P_n(\Theta, \varphi) d\Omega \quad (18)$$

|                                          |                                                |
|------------------------------------------|------------------------------------------------|
| $P_V$                                    | Received spectral power in $\text{W Hz}^{-1}$  |
| $A_e$                                    | Effective area in $\text{m}^2$                 |
| $I_V(\Theta, \varphi)$                   | Intensity in $\text{W Hz}^{-1} \text{sr}^{-1}$ |
| $P_n(\Theta, \varphi)$                   | Normalised power pattern                       |
| $d\Omega = \sin \Theta d\Theta d\varphi$ | Solid angle in sr                              |

If  $I_V = I_c$  is constant, then:

$$P_V = \frac{1}{2} A_e I_c \iint_{\Omega} P_n(\Theta, \varphi) d\Omega. \quad (19)$$

The quantity:

$$\Omega_A = \iint_{2\pi} P_n(\Theta, \varphi) d\Omega \quad (20)$$

is the so-called **beam area**  $\Omega_A$  in unit of sr. It characterises a property of the antenna ('field of view'), also called 'solid angle of the antenna lobe'. This solid angle refers not only to the *main beam*  $\Omega_{MB}$ , but also contains the (unwanted) contribution of the *side lobes*.

Thus, putting all the pieces together, the total power:

$$P = \frac{1}{2} A_e \int_v^{v+\Delta v} \iint_{\Omega} I_v(\Theta, \varphi) P_n(\Theta, \varphi) d\Omega \quad (21)$$

$$= \frac{1}{2} A_e I_c \Omega_A \Delta v. \quad (22)$$

## 2.3 Point-like and extended radio sources, flux density

A source is called ‘point-like’ whose extent is considerably smaller than the lobe width. In the case of less extended sources, especially point-like sources, the radio telescope receives the total intensity emitted over the solid angle  $d\Omega$ . Therefore, instead of the intensity, one often uses the intensity integrated over the solid angle of the source, which is called flux density.

The definition of (spectral) **flux density** is :

$$S_v = \iint_{\Omega_s} I_v(\Theta, \varphi) d\Omega \quad (23)$$

|                        |                                                              |
|------------------------|--------------------------------------------------------------|
| $I_v(\Theta, \varphi)$ | Intensity in $\text{W m}^{-2} \text{Hz}^{-1} \text{sr}^{-1}$ |
| $d\Omega$              | Infinitesimal solid angle in sr                              |
| $S_v$                  | Flux density in $\text{W m}^{-2} \text{Hz}^{-1}$             |
| $\Omega_s$             | Solid angle of the source in sr                              |

The (spectral) power derived in the previous chapter using a radio antenna, can then be represented using flux density instead of intensity.

### 2.3.1 Extended sources

When the antenna is pointed at the centre of an (extended) source, the directional sensitivity of the antenna is taken into account for extended sources:

$$S_v = \iint_{\Omega_s} B(\Theta, \varphi) P_n(\Theta, \varphi) d\Omega. \quad (24)$$

Substituting equation (24) into equation (21), leads to the received power:

$$P = \frac{1}{2} A_e S_v \Delta v. \quad (25)$$

If we also consider  $I_v = I_c$  constant on the antenna lobe, then:

$$S_v = I_c \iint_{\Omega_s} P_n(\Theta, \varphi) d\Omega \approx I_c \Omega_A, \quad (26)$$

with  $\Omega_A$  as beam area. Note that the constraint of integration over the  $\Omega_s$  is equivalent to integration over  $2\pi$  (see definition 20), since  $\Omega_s > \Omega_A$  and  $I_c = 0$  outside  $\Omega_s$ .

The total power is then:

$$P = \frac{1}{2} A_e I_c \Omega_A \Delta\nu. \quad (27)$$

For extended sources one has to integrate over the (main) lobe, the observed flux density  $S_\nu$  is smaller than the true flux density! For extended sources, one therefore often remains with the representation as intensity  $I_\nu$  in  $\text{W m}^{-2} \text{Hz}^{-1} \text{sr}^{-1}$  or as integrated intensity  $I$  in  $\text{W m}^{-2} \text{sr}^{-1}$ .

### 2.3.2 Point-like sources

When the antenna is pointed at a point-like source, then holds  $P_n(\Theta, \varphi) \approx 1$  and:

$$S_\nu \approx \iint_{\Omega_s} I_\nu(\Theta, \varphi) d\Omega. \quad (28)$$

If  $I_\nu = I_c$  is also directionally independent, then:

$$S_\nu = I_c \Omega_s, \quad (29)$$

with  $\Omega_s$  the solid angle of the source.

The solid angle of a distant object with radius  $R$  and distance  $d$  is (see Appendix A):

$$\Omega_s = \frac{\pi R^2}{d^2}. \quad (30)$$

The received power is:

$$P = \frac{1}{2} A_e S_\nu \Delta\nu = \frac{1}{2} A_e I_c \Omega_s \Delta\nu. \quad (31)$$

For point-like sources, therefore, the **flux** in  $\text{W m}^{-2}$  or the **flux density** in  $\text{W m}^{-2} \text{Hz}^{-1}$ , is more often used instead of the intensity  $I_\nu$ .<sup>2</sup>

Comparing the expressions for the received power in the case of extended sources (equation 27) and that of point-like sources (equation 31), the former is limited by the beam area  $\Omega_A$  and the latter by the solid angle of the source  $\Omega_s$ .

## 2.4 Black body radiation

### 2.4.1 Planck's law of radiation

In the 19th century the properties of the radiation of a black body were investigated. The result was the following:

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<sup>2</sup>The terms *intensity*, *flux density* and *flux* are often used incorrectly in the literature. A review of the definition used is usually helpful.

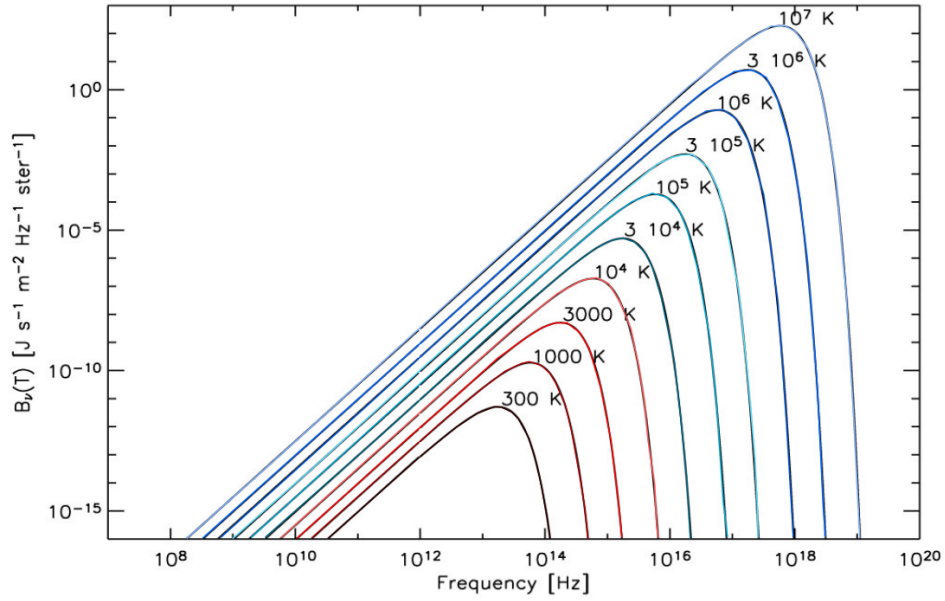


Figure 5: Spectral intensity distribution of black body radiation for temperatures between 300 and 10 million Kelvin.

- The maximum shifts anti-proportionally to the temperature according to **Wien's displacement law**:

$$\lambda_{\max} T = \text{const.}; \quad (32)$$

- According to the **Rayleigh-Jeans law**, the intensity increases with the square of the frequency at low frequencies :

$$B_v \propto \nu^2, \quad (33)$$

or as a function of the wavelength:

$$B_v \propto \lambda^{-2}. \quad (34)$$

- The intensity decreases at high frequencies according to **Wien's radiation law** exponentially:

$$B_v \propto e^{-\nu}; \quad (35)$$

- The total radiation flux increases with  $T^4$  (**Stefan-Boltzmann law**):

$$S \propto T^4 \quad (36)$$

These findings are a consequence of a radiation law found by Max Planck by assuming that the photon energy has discrete values in units of  $h\nu$ , where:

$$h = 6.63 \times 10^{-34} \text{ J s} \quad (37)$$

is the so-called Planck constant.

The Planck function can be written in two forms. First as intensity per wavelength interval:

$$B_{\lambda}(T) = \frac{2hc^2}{\lambda^5} (e^{hc/\lambda kT} - 1)^{-1} \quad (38)$$

$\lambda$  wavelength in m  
 $B_{\lambda}$  intensity in  $\text{W m}^{-2} \text{m}^{-1} \text{sr}^{-1}$   
 $k$  Boltzmann constant  $1.38 \times 10^{-23} \text{ J K}^{-1}$   
 $h$  Planck constant  $6.626 \times 10^{-34} \text{ J s}$   
 $c$  speed of light  $2.99792 \times 10^8 \text{ m s}^{-1}$

Second as intensity per frequency interval:

$$B_{\nu}(T) = \frac{2h\nu^3}{c^2} (e^{h\nu/kT} - 1)^{-1} \quad (39)$$

$B_{\nu}$  intensity in  $\text{W m}^{-2} \text{Hz}^{-1} \text{sr}^{-1}$   
 $\nu$  frequency in Hz

The behaviour of the intensity as a function of the temperature in frequency notation is shown in Fig. 5.

The last two forms can be transformed into each other by

$$B_{\lambda} d\lambda = -B_{\nu} d\nu. \quad (40)$$

using  $\nu = c/\lambda$  and  $d\nu = (-c/\lambda^2)d\lambda$

#### 2.4.2 Wien's displacement law

Wien's displacement law follows finding the maximum of the Planck function by setting its derivative to zero. From  $e^{h\nu/kT} \gg 1$ , it follows:

$$\frac{h\nu_{\max}}{kT} = 3, \quad (41)$$

and thus:

$$\lambda_{\max} T = \frac{hc}{3k} = 0.0048 \text{ m K}, \quad (42)$$

where  $\nu_{\max}$  and  $\lambda_{\max}$  are the frequency and wavelength corresponding to the maximum of Planck curve. This is valid for Planck function  $B_{\nu}$  related to a frequency interval. For  $B_{\lambda}$  these maxima are reached at a different wavelength (frequency):

$$\lambda_{\max} T = 0.00288 \text{ m K}. \quad (43)$$

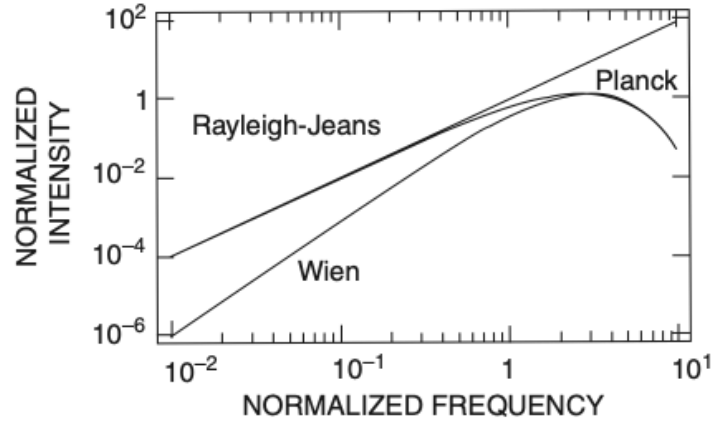


Figure 6: Schematic intensity distribution of the Planck law and its approximations. From Wilson (2008).

### 2.4.3 Rayleigh–Jeans law

The Rayleigh-Jeans law holds for  $h\nu \ll kT$ . It is obtained by developing  $e^{h\nu/kT}$  into a series:

$$B_\nu = \frac{2h\nu^3}{c^2} \frac{kT}{h\nu} = \frac{2kT}{\lambda^2}. \quad (44)$$

**The Rayleigh–Jeans law (RJ) is fundamental for radio astronomy, because in the radio range ( $\lambda > 1$  mm),  $h\nu \ll kT$  holds for all realistic temperatures, and thus the Rayleigh-Jeans approximation is fulfilled.**

### 2.4.4 Wien’s law of thermal radiation

Wien’s radiation law is valid for  $e^{h\nu/kT} \gg 1$ :

$$B_\nu = \frac{2h\nu^3}{c^2} e^{-\frac{h\nu}{kT}}. \quad (45)$$

The schematic behaviour of the intensity as a function of the temperature for the Rayleigh–Jeans law and Wien’s radiation law is shown in Fig. 6.

### 2.4.5 Stefan–Boltzmann Law

The total intensity is obtained by integration over the Planck curve:

$$B = \frac{2h}{c^2} \int_0^\infty \frac{\nu^3}{e^{h\nu/kT} - 1} d\nu, \quad (46)$$

The Stefan-Boltzmann law is then:

$$B = \sigma' T^4, \quad (47)$$

with  $\sigma' = 1.8 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4} \text{ sr}^{-1}$ .

If one drops the directional dependence of the intensity, one obtains the total radiation flux (cf. equation 15):

$$S = \pi B, \quad (48)$$

and thus:

$$S = \sigma T^4, \quad (49)$$

with  $\sigma = \pi\sigma' = 5.669 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$  as Stefan–Boltzmann constant.

## 2.5 Radiation temperature and flux density

A central measurement of radio astronomy is the **radiation temperature** or brightness temperature  $T_b$ . It is closely related to the **flux density**  $S_\nu$ , which is often used in its place.

The Rayleigh–Jeans law (equation 44) applies to the intensity::

$$B_\nu = \frac{2kT_b}{\lambda^2}. \quad (50)$$

Thus, the flux density can be expressed by equation (29) as:

$$S_\nu = B_\nu \Omega_s. \quad (51)$$

For point-like sources then holds:

$$S_\nu = \frac{2kT_b\Omega_s}{\lambda^2}. \quad (52)$$

**In honour of the discoverer of radio radiation from space, Karl G. Jansky, the unit Jansky (Jy) is used in radio astronomy :**

$$1 \text{ Jy} = 10^{-26} \text{ W m}^{-2} \text{ Hz}^{-1}.$$

## 3 Parabolic antennae

### 3.1 Construction

For single-dish antennae, the parabolic antenna has become the accepted design. Horn antennas are preferably used for the absolute calibration of the radiation flux. Freely movable parabolic antennae have been constructed with diameters of up to  $\approx 100 \text{ m}$ .

Good focusing in any position of the telescope is not realisable with a rigid construction. With the technical principle of **homologous deformation**, the antenna deforms under its own weight and changes back to a parabolic shape with a different focus. The focusing must therefore be changed when the position of the telescope is changed.



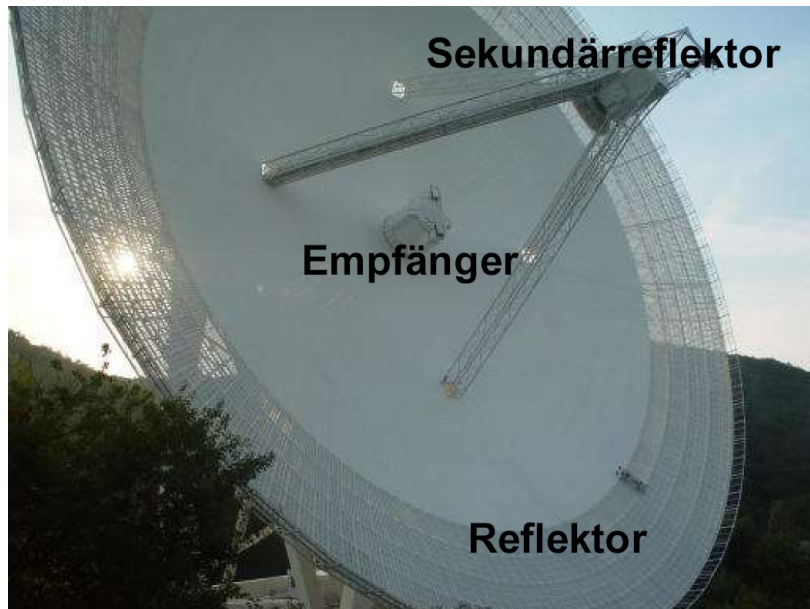


Figure 7: 100-m Effelsberg radio telescope. The radiation received by the reflector is directed via the secondary reflector into the receiver cabin. There, various receivers for different frequency ranges can be used. As an alternative to the secondary reflector, a further receiver cabin at the top of the mast construction can be used (primary focus).

The **azimuthal mount** is preferred for radio antennae. The disadvantage of this design compared to the **parallactic mount** is the field rotation during tracking. As coordinates of the telescope system, azimuth and elevation are used.

Radiation at radio frequencies is reflected from metal surfaces practically without loss if the unevenness is smaller than  $\lambda/10$ , better still  $\lambda/20$ . This means that in the range of cm- to m-waves the construction with wire grids or perforated panels is preferred. This saves weight and thus costs for the mounting. Also the wind load on the telescope is effectively reduced.

Even in the mm range, the accuracy of the antennas is only  $\leq 0.1$  mm. For comparison, an optical telescope has a mirror accuracy of only  $\approx 0.02 \mu\text{m}$ .

### 3.2 Elements of a receiving system

The elements of the radio receiving system consist of the main mirror, the subreflector or primary focus cabin, the horn (feed antenna) and the receiver (RX) (Fig. 7).

The ‘optical systems’ of radio antennae are designed differently, for instance with regards to the choice of the location of the receiver. The primary focus has advantages at low frequencies, but the change of the receiver is cumbersome. The secondary focus can usually accommodate several receivers which can be flexibly switched between during an observation.

### 3.2.1 Aperture efficiency

Every antenna with a geometric area  $A$  has losses, i.e. it receives as much power as a lossless antenna with a (smaller) area  $A_e$ . The **aperture efficiency**  $\eta$  indicates how high the losses are:

$$A_e = \eta A, \quad (53)$$

with

- $A_e$  = effective antenna area in  $\text{m}^2$
- $A$  = geometric antenna area in  $\text{m}^2$
- $\eta$  = aperture efficiency

The aperture efficiency  $\eta$  is a product of several loss factors:

$$\eta = \eta_{\text{SF}} \eta_{\text{BL}} \eta_s \eta_t \eta_{\text{misc}} \quad (54)$$

with

- $\eta_{\text{SF}}$  = surface accuracy ( $\approx 0.5$ );  $\eta_{\text{SF}}^{\text{min}}$  at the highest frequency
- $\eta_{\text{BL}}$  = shading by subreflector and supports  $0.75 \leq \eta_{\text{BL}} \leq 0.95$
- $\eta_s$  = losses at the feed horn  $0.7 \leq \eta_s \leq 0.97$
- $\eta_t$  = uneven illumination by the feed horn  $\eta_t \geq 0.9$
- $\eta_{\text{misc}}$  = other losses  $\eta_{\text{misc}} \geq 0.95$

Typical values for the aperture efficiency are  $\eta \approx 0.4\text{--}0.6$ .

### 3.3 Antenna temperature, noise temperature and Nyquist theorem

If one connects an amplifier with a resistor  $R$  and the temperature  $T$ , then this measures a power dependent on the temperature. The amplifier creates noises according to its so-called **noise temperature**

The thermal motion in the resistor produces a current  $i(t)$  which is zero averaged over a long period of time ( $\langle i(t) \rangle = 0$ ).

However,  $\langle i^2 \rangle > 0$ . Since  $i^2$  corresponds to a power, the resistor transfers power to the amplifier. Nyquist (1929) shows that the process corresponds to the random walk process of Brownian motion and the relation holds:

$$p = kT \quad (55)$$

with

- $p$  = spectral power in  $\text{WHz}^{-1}$
- $k$  = Boltzmann constant
- $T$  = temperature in K

For a power over a bandwidth  $\Delta\nu$ ,  $P = kT\Delta\nu$ , with  $P$  in Watt.

If the amplifier measures a power  $P$  at a radio receiving antenna then this corresponds to an **antenna temperature**  $T_A$  of:

$$P = kT_A\Delta\nu. \quad (56)$$

The antenna temperature  $T_A$  is therefore another measure with which the received power can be described. In radio astronomy  $T_A$  is used frequently.

According to equation (22) the power received for **extended sources**:

$$P = \frac{1}{2}A_e B_c \Omega_A \Delta\nu \quad (57)$$

with

- $A_e$  = antenna effective area in  $\text{m}^2$
- $B_c$  = intensity (again, assumed to be constant) in  $\text{W m}^{-2} \text{Hz}^{-1} \text{sr}^{-1}$
- $\Omega_A$  = beam area in sr
- $\Delta\nu$  = bandwidth in Hz

By equating equations (56) and (57) and using Rayleigh–Jeans law ( $B_c = 2kT_b/\lambda^2$ ), one obtains:

$$T_A = \frac{A_e \Omega_A}{\lambda^2} T_b \quad (58)$$

with  $T_b$  as radiation temperature.

For antennae, the **antenna theorem** applies in general:

$$A_e \Omega_A = \lambda^2. \quad (59)$$

The theorem means that a large effective antenna area  $A_e$  necessarily implies a narrow antenna lobe. It is not possible to build instruments with a wide antenna lobe that do not at the same time significantly lose sensitivity. Note that  $\Omega_A$  is only approximately the solid angle of the main lobe  $\Omega_{MB}$ , but also contains the contribution of the sidelobes.

It follows a heuristic derivation of the antenna theorem. The resolution of a telescope, i.e. the half width  $q$  of the antenna lobe is  $q \approx \lambda/D$ , so:

$$\Omega_A \propto q^2 \approx \frac{\lambda^2}{D^2} \quad (60)$$

But since  $A_e \propto D^2$ , it follows that  $\Omega_A \propto \lambda^2/A_e$ . The exact calculation gives  $\Omega_A A_e = \lambda^2$ . Using equation (59), it follows from equation (58) for extended sources:

$$T_A = T_b, \quad (61)$$

i.e. the measured antenna temperature corresponds to the brightness temperature of the source.

For **point-like sources**, equation (31) applies:

$$P = \frac{1}{2}A_e S_\nu \Delta\nu = \frac{1}{2}A_e B_c \Omega_s \Delta\nu \quad (62)$$

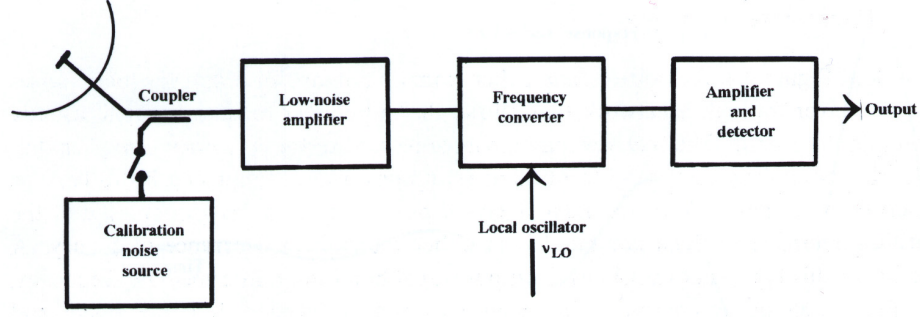


Figure 8: Schematic structure of a radio receiver system.

and from it after appropriate calculation:

$$T_A = \frac{\Omega_s}{\Omega_A} T_b \quad (63)$$

i.e.  $T_A$  becomes smaller for a given  $T_b$ , the larger  $\Omega_A$  (the antenna lobe) becomes. This effect is also called the **dilution effect**.

### 3.3.1 Antenna temperature and flux density

For point-like sources, the flux density  $S_\nu$  is calculated from the antenna temperature  $T_A$  by making use of the aperture efficiency  $\eta$ . According to equations (52) and (63), the antenna theorem and the definition of the aperture efficiency  $\eta$  (equation 53) holds:

$$S_\nu = \frac{2kT_b\Omega_s}{\lambda^2} = \frac{2kT_A\Omega_A}{\lambda^2} = \frac{2kT_A}{A_e} = \frac{2k}{\eta A} T_A \quad (64)$$

The assumption of aperture efficiency to estimate flux density is often used in mm and sub-mm astronomy when suitable flux calibrators are not available or have not been measured.

## 4 Radio receiver systems, calibration and observing technique

The path of the received signal through the telescope and the receiving system is shown schematically in the block diagram of Fig. 8. In principle, the system contains the following components:

- Antenna with receiving horn;
- First amplification level;
- Mixer;

- Further amplification stages and the detector.

After signal amplification in the first amplification stage, the signal is converted to another frequency in a mixer. The signal is then further amplified and fed to a detector.

The receiver arrangement at the secondary focus of the Effelsberg radio telescope is shown in Fig. 9.

## 4.1 Receiver types and low noise amplifiers

The receiver designations depend on the technology used for the first amplifier stage. Today, the receivers used are:

In the cm range up to  $\approx 100$  GHz: Cooled High Electron Mobility Transistors (HEMT)  
 $> 100$  GHz (mm range): SIS-Mixer (Superconductor - Insulator - Superconductor)

The first amplifier stage must consist of so-called **low-noise amplifiers (LNA)**, since this dominates the noise budget of all amplifier stages together. According to the formula of H. T. Friis, the total noise of the system is  $P_{\text{total}}$ :

$$P_{\text{Gesamt}} = P_{\text{LNA}} + \frac{P_{\text{further}}}{G_{\text{LNA}}} \quad (65)$$

where,  $P_{\text{LNA}}$  is the noise of the LNA,  $P_{\text{further}}$  the noise of all further stages and  $G_{\text{LNA}}$  is the gain factor of the LNA. The LNA must therefore have as high a gain as possible and lowest possible inherent noise. The cooled HEMT amplifiers are standard today and are a special type of field effect transistor amplifier.

### 4.1.1 System temperature

The antenna with feed horn, the cable connection to the LNA and the amplifier itself contribute to the total noise of the receiving system. The noise of the stages after the LNA can be neglected. The LNA is placed immediately after the feed horn to minimize line losses. The noise temperature describing the total noise (cf. Chapter 3.3) is also called **system temperature**  $T_{\text{sys}}$ .

Example: System temperature at 5 GHz ( $\lambda \approx 6$  cm)

|                                          |                       |
|------------------------------------------|-----------------------|
| Contribution of the antenna:             | $T = 15\text{--}20$ K |
| Contribution of LNA (HEMT):              | $T = 20\text{--}30$ K |
| $T_{\text{sys}} \approx 30\text{--}50$ K |                       |

### 4.1.2 Amplification factor

The system temperature corresponds to a spectral power:  $P_v = kT_{\text{sys}}$  and a total power  $P = kT_{\text{sys}} \Delta\nu$ . With  $\Delta\nu \approx 50\text{--}100$  MHz, we thus find  $P = 5 \times 10^{-14}$  W.

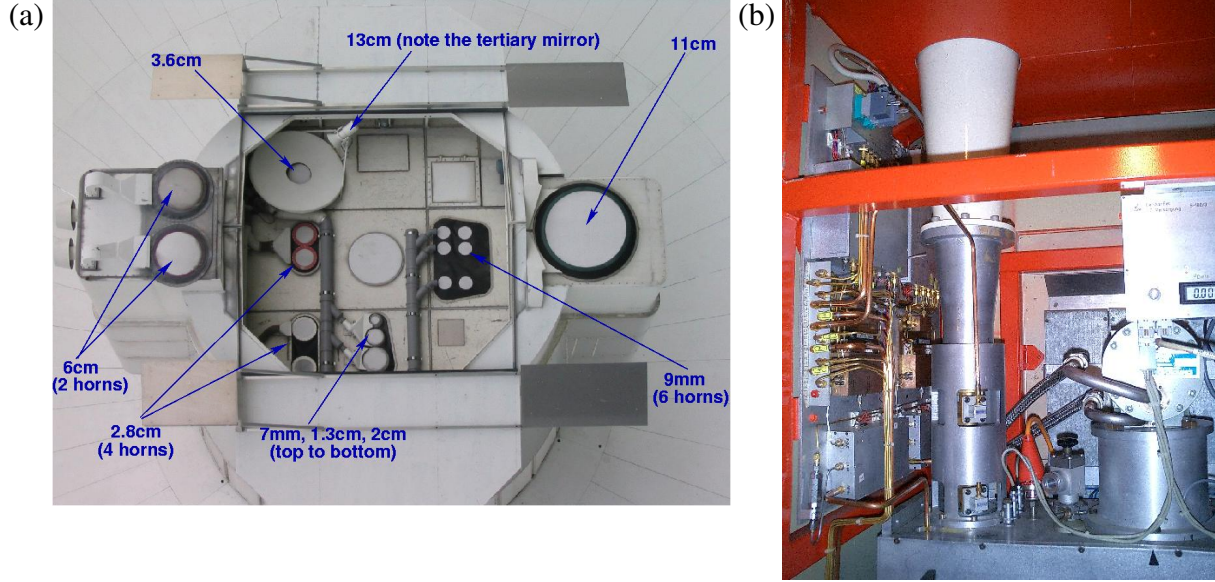


Figure 9: (a) Arrangement of the receiving horns in the secondary focus cabin at Effelsberg. (b) Example receiving horn with receiver electronics.

The weak radio signals are of the same strength and must be amplified in order to produce a power of  $\approx 10$  mW at the detector, i.e. a gain of  $\approx 5 \times 10^{12}$  is required.

In communications engineering, the amplification factor  $G$  is usually given in decibels (dB). Thus, a gain of  $G = 5 \times 10^{12}$  also corresponds to  $G = 127$  dB, where  $1 \text{ dB} = 10 \log_{10}(P_{\text{out}}/P_{\text{in}})$  with the input power  $P_{\text{in}}$  and the output power  $P_{\text{out}}$ .

The weakest detectable signal for a given system temperature  $T_{\text{sys}}$  is given by the radiometer equation (82). According to this equation with  $T_{\text{sys}} = 50$  K the smallest detectable temperature is  $\approx 5$  mK ( $\Delta\nu = 100$  MHz,  $\tau = 1$  s), so that a high stability of the receiving system is required, typically 1/10000.

## 4.2 Mixers and backends (detectors)

Usually the receiving system contains a **mixer** which transfers the observed frequency band centred on the frequency  $\nu_0$  into another frequency band without changing the signal as such (so-called **heterodyne principle**). The reason is a better further processing of the signal. Another reason is the reduction of feedback to the input signal. The mixer separates the receiving system into the ‘**front-end**’ area and the ‘**back-end**’ area. The front-end range is specific for the frequency band (horn, LNA), whereas the back-end range can be used for different frequency bands after frequency conversion. Frequency transmission is performed by injecting a sinusoidal signal of frequency  $\nu_{\text{LO}}$  through a **local oscillator (LO)**. The resulting frequency  $\nu_{\text{M}}$  is the difference or sum of the two input frequencies:

$$\nu_{\text{M}} = \nu_0 \pm \nu_{\text{LO}} . \quad (66)$$

The two newly generated frequency bands are called **upper side band (USB)** or **lower side band (LSB)**. Usually only the signal of one of the two bands is processed.

In the back-end area, three types of devices are used, depending on the application:

- Continuum measurement (Continuum back-end);
- Spectral lines (Spectrometer);
- Time resolution (Pulsar back-end).

Spectrometers are divided into filter spectrometers, auto-correlation spectrometers and acoustic-optical spectrometers.

A filter spectrometer is obtained by constructing  $n$  parallel continuum backends with a bandwidth of  $\Delta\nu = n\delta\nu$ , where  $\delta\nu$  is the spectral resolution. This setup is typical for resolutions in the kHz-range.

Modern auto-correlation spectrometers allow flexible use by selecting the specific resolution.

The Acoustic-Optical Spectrometer offers advantages especially at high bandwidths.

#### 4.2.1 Example: Use of the Effelsberg 1024-channel auto-correlator

A stellar maser with lines in an expected velocity interval of  $40 \text{ km s}^{-1}$  is measured. The autocorrelator allows bandwidths  $\Delta\nu$  of 3.125, 6.25, 12.5 and 25 MHz. The covered velocity interval  $\Delta v$  is obtained approximately from:

$$\frac{\Delta v}{c} = \frac{\Delta\nu}{\nu_0}, \quad (67)$$

with  $\nu_0 = 22235.08 \text{ MHz}$  (rest frequency of the water vapour maser line),  $c$  = speed of light. If  $\Delta\nu = 3.125 \text{ MHz}$  is chosen,  $\Delta v = 42 \text{ km s}^{-1}$ , and thus the expected velocity interval is covered only just. Therefore,  $\Delta\nu = 6.25 \text{ MHz}$  is chosen instead to be on the safe side.

The velocity resolution  $\delta v$  results from:

$$\frac{\delta v}{c} = \frac{\delta\nu}{\nu_0}, \quad (68)$$

where  $\delta\nu = \Delta\nu/1024$ . For  $\Delta\nu = 6.25 \text{ MHz}$ , one obtains  $\delta\nu = 6.1 \text{ kHz}$  and  $\delta v = 0.08 \text{ km/s}$ !

## 4.3 Calibration

### 4.3.1 Earth atmosphere, background noise

Every radio measurement is limited by noise, which is composed of contributions from the background (atmosphere, galactic background) and the receiving system. Thus, for the measured antenna temperature:

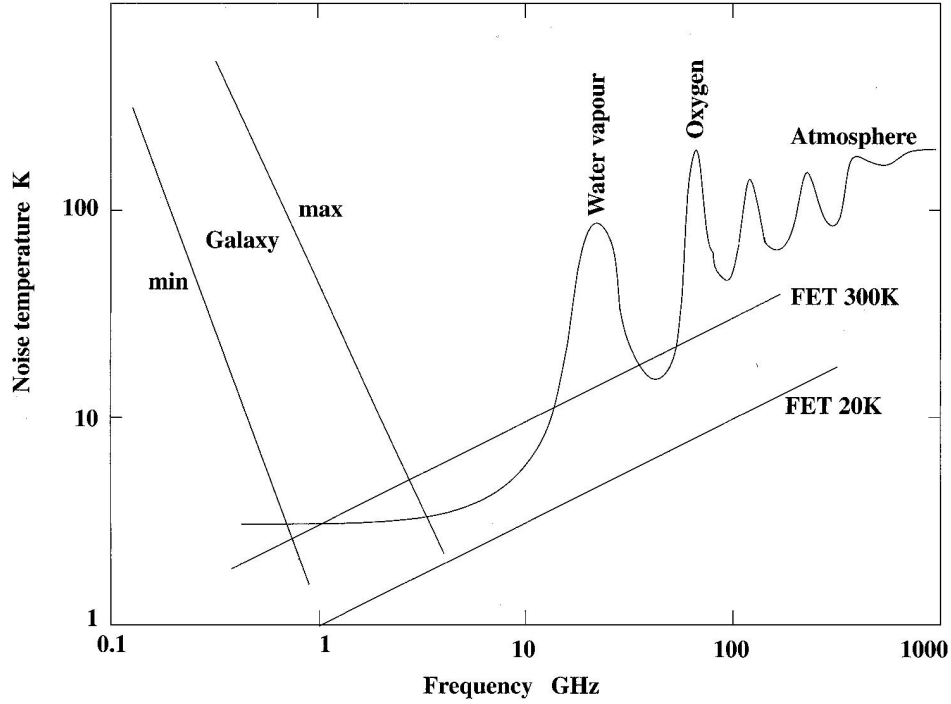


Figure 10: Noise temperatures of the sky background and of two receivers. The lowest sky background is observed between 1 and 10 GHz. There, cooled receivers (FET 20 K) are used.

$$T_A = T_s + T_h + T_{\text{sys}} \quad (69)$$

with

- $T$  = measured antenna temperature
- $T_s$  = temperature of the source
- $T_h$  = temperature of the background
- $T_{\text{sys}}$  = temperature of the receiving system

$T_h$  contains components from the atmosphere and scattering contributions from the side lobes. The task of the receiving system is to separate  $T_s$  from  $T_h$  and  $T_{\text{sys}}$ .  $T_{\text{sys}}$  contains parts of the antenna and the amplifier.

Die Fig. 10 shows typical **noise temperatures for the background** and for common receivers. At low frequencies up to about 2 GHz, the Galactic background noise (synchrotron radiation from electrons in the Galactic magnetic field) dominates. In the cm range, the atmosphere and scattered radiation from the ground dominate. A strong increase of the background occurs in the mm-wavelength range by radiation from water vapor and oxygen (with peaks at 22 GHz and 55–60 GHz).

For comparison, the noise temperature of typical HEMT receivers with a FET (field effect transistor) is plotted.



### 4.3.2 Calibration of the system temperature

The system temperature can be determined by feeding a noise signal from a resistor whose temperature is precisely known. This can be done using either a sampler of a given temperature or a noise diode.

The temperature of the receiving system (in this case feed horn, amplifier, their interconnection)  $T_{\text{sys}}$  can be determined by the so-called **hot-cold method**.

If a warm absorber (e.g. with ambient temperature  $T_h$ ) is held in front of the feed horn, the receiver measures the power:

$$P_h = c k (T_h + T_{\text{sys}}) \Delta \nu , \quad (70)$$

where  $c$  is a proportionality constant,  $k$  is the Boltzmann constant and  $\Delta \nu$  is the bandwidth of the receiver. If the absorber is then cooled, e.g. to the temperature of liquid nitrogen ( $T_c = 78 \text{ K}$ ), the receiver measures the power:

$$P_c = c k (T_c + T_{\text{sys}}) \Delta \nu . \quad (71)$$

Defining  $y = P_h/P_c$ , the temperature of the receiving system can be determined from:

$$T_{\text{sys}} = \frac{T_h - y T_c}{y - 1} . \quad (72)$$

This calibration is rarely performed because of the effort involved.

The temperature  $T_{\text{cal}}$  of a noise diode can also be determined by means of the hot-cold method. With a given temperature stability, this can then be used as a calibration source. Since this does not block the sky, the following applies:

$$T'_{\text{sys}} = T_{\text{sys}} + T_h , \quad (73)$$

so that the system temperature here also contains the term  $T_h$  from equation (69). The measured power with the diode switched on is then:

$$P_{\text{on}} = c k (T_{\text{cal}} + T'_{\text{sys}}) \Delta \nu , \quad (74)$$

and with the diode switched off:

$$P_{\text{off}} = c k T'_{\text{sys}} \Delta \nu . \quad (75)$$

In this case, one obtains  $T'_{\text{sys}}$  from:

$$T'_{\text{sys}} = T_{\text{cal}} \frac{P_{\text{off}}}{P_{\text{on}} - P_{\text{off}}} . \quad (76)$$

Alternatively, calibration sources can be measured in the sky, which, however, must first be calibrated by comparison with a noise source. These calibration sources must also be monitored regularly to ensure that they are not subject to change.

Lists of calibration sources are kept at the major radio observatories. Important calibration sources are 3C 286 (quasar) or NGC 7027 (planetary nebula).

## 4.4 Sensitivity limit

The uncertainty  $\Delta T_A$ , with which the receiver can measure the temperature  $T_A$  depends on the bandwidth  $\Delta \nu$  and the integration time  $\tau$  as:

$$\Delta T_A = \frac{C_s T_A}{\sqrt{\Delta \nu \tau}}, \quad (77)$$

where  $C_s$  is a correction factor ( $C_s \approx 1$ ).  $\Delta T_A$  is the minimum temperature difference that can still be measured.

Equation (77) is derived from the following considerations. For a given bandwidth  $\Delta \nu$  centred on the frequency  $\nu_0$ , the energy of the photons registered in the receiver lies in the interval  $[\nu_0 - \Delta \nu/2, \nu_0 + \Delta \nu/2]$ , i.e. the accuracy with which the energy of the photon can be determined is  $\Delta E = h \Delta \nu$  ( $h$  = Planck's constant). According to Heisenberg's uncertainty relation  $\Delta E \Delta t \geq h$  applies, i.e. the minimum time  $\Delta t$  for the measurement of the photon amounts to  $\Delta t = h/h \Delta \nu = \Delta \nu^{-1}$ . Example, for  $\Delta \nu = 3$  kHz is  $\Delta t = 10^{-3}$  s. Therefore, the receiver generates per second  $\Delta \nu$  statistically independent noise contributions.

Within the integration time  $\tau$ , one thus has  $n = \Delta \nu \tau$  independent contributions. In fact one measures  $n \pm \Delta n$  contributions within  $\tau$ . The uncertainty is described by Poisson noise, i.e.  $\Delta n = \sqrt{n}$ .

If a measure, such as the antenna temperature  $T_A$  is proportional to  $n$ , then is valid:

$$\frac{\Delta T_A}{T_A} = \frac{\Delta n}{n} = \frac{\sqrt{n}}{n} = \frac{1}{\sqrt{n}} = \frac{1}{\sqrt{\Delta \nu \tau}}, \quad (78)$$

and thus equation (77), except for the instrumental factor.

For a given antenna temperature  $T_A$  the minimum temperature difference that can still be measured is  $\Delta T_A$  is given by equation (77). This can be converted into a flux density difference by the Rayleigh-Jeans law:

$$\Delta S_\nu = \frac{2k \Delta T_A}{\lambda^2} \Omega_A \quad (79)$$

By substituting equation (77) one obtains a minimum flux density difference  $\Delta F$  of:

$$\Delta S_\nu = \frac{2k}{\lambda^2} \frac{C_s T_A}{\sqrt{\Delta \nu \tau}} \Omega_A. \quad (80)$$

The solid angle of the antenna lobe  $\Omega_A$  is determined by the wavelength and the effective antenna area  $A_e$  determined by  $A_e \Omega_A = \lambda^2$  (equation 59), thus:

$$\Delta S_v = \frac{2k}{A_e} \frac{C_s T_A}{\sqrt{\Delta \nu \tau}}. \quad (81)$$

According to equation (69),  $T_A$  consists of several components. In general, the source contribution is small compared to the background and instrumental contributions. Usually  $T_A = T'_{\text{sys}}$  is assumed, referring either here to the instrumental contribution alone or also additionally to the background contribution.

The result is the so-called **Radiometer-equation**:

$$\Delta S_v = \frac{2k}{A_e} \frac{C_s T'_{\text{sys}}}{\sqrt{\Delta \nu \tau}} \quad (82)$$

One may ask what is the theoretical lower limit for the noise in a receiver, i.e. what is the **minimum possible noise temperature**  $T_{\text{min}}$  is? According to the rules of quantum mechanics,  $kT_{\text{min}} = h\nu$ . For example, for  $\nu = 5$  GHz,  $T_{\text{min}} = 0.24$  K and for  $\lambda = 2.6$  mm,  $T_{\text{min}} = 5.5$  K. Real receiving systems are far from this limit, e.g.  $T_{\text{sys}} \approx 50$  K in cm-range. The noise temperatures of the background, e.g. in the metre and millimetre range, are also much higher (cf. Fig. 10).

#### 4.4.1 Signal-to-noise ratio

The ratio  $T_A/\Delta T_A$  is called the **signal-to-noise ratio (S/N)**. Good quality measurements should have  $S/N = 5$ .

Excellent measurements are characterized by  $S/N > 10$ .

Note: A doubling of the measurement quality (the S/N) means a quadrupling of the integration time, i.e. reducing the measurement limit from 0.5 Jy to 0.1 Jy means 25 times the integration time. Correspondingly, a halving of the target limit sensitivity means a quadrupling of the observation time.

### 4.5 Observation technique

The more accurate a measurement is to be, the more often control measurements are necessary. These control measurements are necessary for the alignment accuracy as well as for the flux calibration (see chapter 4.3.2). In addition, there is time for focusing the telescope. The focus must be re-checked in case of significant temperature changes, e.g. during dawn. The total time necessary for this is called **overhead time** and depends on the total available observation time, i.e. the total available observation time  $T_G =$  scientifically usable time  $T_w$  + overhead  $T_k$ .

$$\begin{aligned} T_G &= T_w + T_k \\ T_w &= \text{integration time on source under investigation + slewing time between sources} \\ T_k &= \text{flux calibration measurements + pointing control +} \\ &\quad \text{determination of the atmospheric opacity curve} \end{aligned}$$

Typically 20 percent of the available observation time is used for calibration measurements to control instrumental and weather dependent effects. The more unstable the receiving system or the weather conditions, the more often such measurements are necessary.

#### 4.5.1 Pointing accuracy

Let  $\Delta\Theta$  be the error of the antenna pointing (e.g. in arcsec) and  $\Theta_{3\text{dB}}$  be the width of the lobe at a sensitivity drop of 3 dB (about a factor of 2) then the desired pointing accuracy is:

$$\Delta\Theta < \Theta_{3\text{dB}}/20 . \quad (83)$$

This keeps errors in flux density measurements  $<1\%$ . The typical pointing errors for large radio telescopes are in the order of 10 arc seconds. However, the pointing accuracy depends on the telescope position, so that the nominal position has to be approached with the help of an empirical correction model ( $\Delta\text{Azimut}$ ,  $\Delta\text{Elevation}$ ).

Problems: The wind and temperature dependence cannot be calibrated, so that regular control measurements are necessary.

Control measurements are realised by continuum scans in azimuth and elevation. The telescope passes over a strong point source in each direction. The flux density maximum must be measured in both directions at the nominal position, otherwise a manual correction is necessary.

#### 4.5.2 Focus

At the beginning of an observation, the focus of the telescope must be optimized. For this purpose, the flux density of a strong source is measured as a function of the distance of the secondary mirror (or the primary focus receiver) from the primary mirror. At optimum distance, the flux density is at a maximum. Re-focusing during the observation becomes necessary when the outside temperature changes significantly, e.g. after sunrise.

#### 4.5.3 Calibration source measurement for flux calibration

Since the receiving characteristics of the antenna and the transmission of the atmosphere change with the distance to the zenith ( $= 1 - \text{elevation of the antenna}$ ), calibration sources for flux densities must be measured over the relevant elevation range to determine the elevation-dependent corrections.

#### 4.5.4 Velocity corrections

In the case of spectral line observations, a source with lines of known velocities shall also be included in order to check the correctness of the velocity corrections automatically made in the computer.

To determine the velocity of a spectral line with respect to a reference location, the velocity of the receiving system with respect to this reference location must be calculated. The rotation of the earth and the movement of the earth with respect to the reference location contribute to this velocity.

The velocities are then given in the **heliocentric system**. If one corrects the velocity of the solar system centre of gravity by the relative velocity of the Sun with respect to the surrounding gas, one obtains velocities relative to the **local standard of rest (LSR)**. This location coincides with the location of the Sun and moves with the local orbital velocity around the centre of the Milky Way.

Radio astronomical velocities are usually given relative to the LSR.

#### 4.5.5 Definition of Doppler velocities

There are two possibilities to assign a radial velocity to a frequency. The radial definition is:

$$v_{\text{rad}} = c \frac{v_0 - \nu}{v_0}, \quad (84)$$

where  $c$  = speed of light,  $v_0$  = rest frequency,  $\nu$  = observed frequency, and  $v_{\text{rad}}$  = radial velocity.

An alternative definition (optical definition) is:

$$v_{\text{rad}}^{\text{opt}} = c \frac{\lambda - \lambda_0}{\lambda_0}, \quad (85)$$

where  $c$  = speed of light,  $\lambda_0$  = rest wavelength,  $\lambda$  = observed wavelength, and  $v_{\text{rad}}^{\text{opt}}$  = radial velocity.

The difference  $\Delta v$  in velocities for the two definitions is:

$$\Delta v = c \frac{(v_0 - \nu)^2}{v_0 \nu}. \quad (86)$$

The difference is minimal for velocities occurring in the Milky Way, but becomes important for extra-galactic velocities. The optical definition is the official definition, but the radio definition is often used in radio astronomy for practical reasons.

#### 4.5.6 Position-, beam- and frequency-switching

The limit sensitivity calculations (Section 4.4) apply to pure ON-measurements. Due to the instabilities, especially in the atmosphere, differential measurements have to be made, i.e. for each recording with source (ON) a second one without (OFF) is made. There are different methods to realise this. With the **position switching** method the OFF measurement is made after moving the telescope to a position with distance three times the lobe width to the ON position. In the **beam switching** method, there are two parallel receiving systems of which one

is directed to the source and one to the sky. This method is twice as fast as position switching. After a measurement cycle, the source is placed in the second receiving system.

**Frequency switching** is used when spectral lines are observed. In this case the receiver frequency is changed during the OFF measurement, so that the measurement does not contain the line to be observed. It is also possible to place the line in the observed band at different positions, so that the same speed as with beam switching is achieved. Depending on the observation method, the limit sensitivity changes per time unit of observation. With the KRT 3, only position switching, also called **ON-OFF measurement**, is used.

## 5 The Sun as a radio star

The radio waves of the Sun were discovered in 1942 during a ‘radio storm’ by British radar ( $\lambda 4-8$  m). This disturbance was about  $10^6$  times stronger than the expected thermal radiation of the Sun.

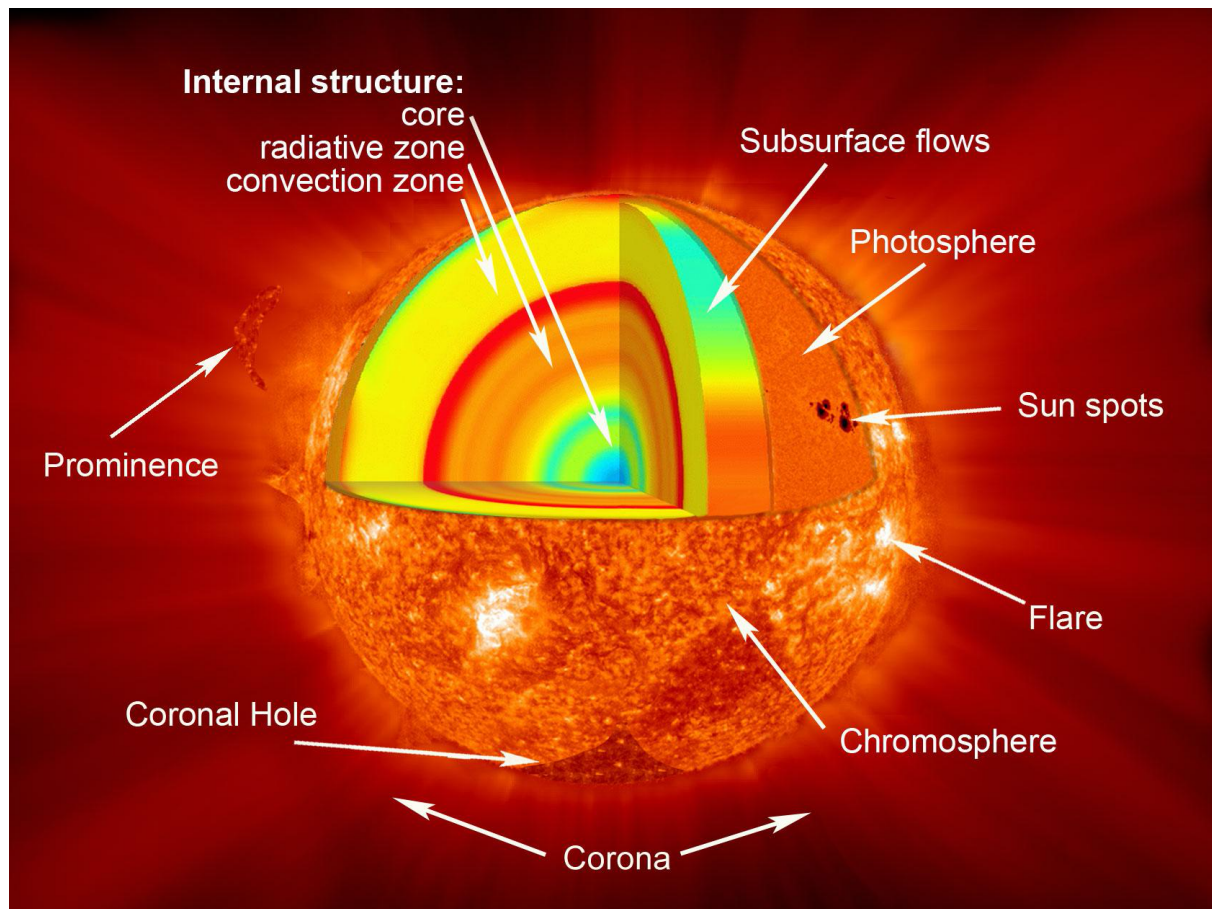


Figure 11: Schematic structure of the Sun. *Source: NASA.*

## 5.1 The structure of the Sun

Basic parameters::

- Mass:  $M_{\odot} = 1.98 \times 10^{30}$  kg,  $M_{\odot} = 332.800 M_E$  ( $M_E$  = Earth's mass), 345 x sum of all planetary masses;
- Radius:  $R_{\odot} = 700.000$  km,  $108 R_E$  ( $R_E$  = Earth's radius);
- Surface temperature (photosphere):  $T_{\odot} = 5780$  K;
- Luminosity intensity:  $L_{\odot} = 3.83 \times 10^{26}$  W.

Fig. 11 shows the basic structure of the Sun schematically. The Sun consists of several components:

- The core: energy generation  $r_K \approx 0.3 R_{\odot}$  ( $r_K$  = core radius)  
 $V_K \approx 0.1 V_{\odot}$ , but  $M_K = 0.4 M_{\odot}$  ( $V_K$  = core volume);
- The mantle: convection currents in the upper layers  $T_R \approx 7 \times 10^6$  K  
 $T_R$  = temperature at the core-shell transition;
- The photosphere: 'surface' of the Sun layer of high opacity, some 100 km thick; absorption lines give chemical composition, centre-rim darkening, granulation some 100 km large;
  - Sunspots: slightly cooler temperature than surroundings. Disturbances from deeper layers. Magnetic fields with field strengths of  $\approx 4000$  Gauss;
- The chromosphere: about 6000 km thick, temperature  $\approx 7000$  K. Glows red because of  $H\alpha$ - emission;
- The corona: extension of several solar radii. Temperature  $2 \times 10^6$  K with hot spots  $5 \times 10^6$  K. Very thin gas (plasma), the temperature corresponds to the kinetic temperature;
- More details: J.B. Kaler, Stars, Freeman N.Y. 2002, p. 107 f.

## 5.2 Radio components of the Sun

The radio radiation of the Sun consists of different components:

- the quiet Sun;
- the disturbed Sun;
- the slowly variable component: time scale  $t > \text{days}$ ;

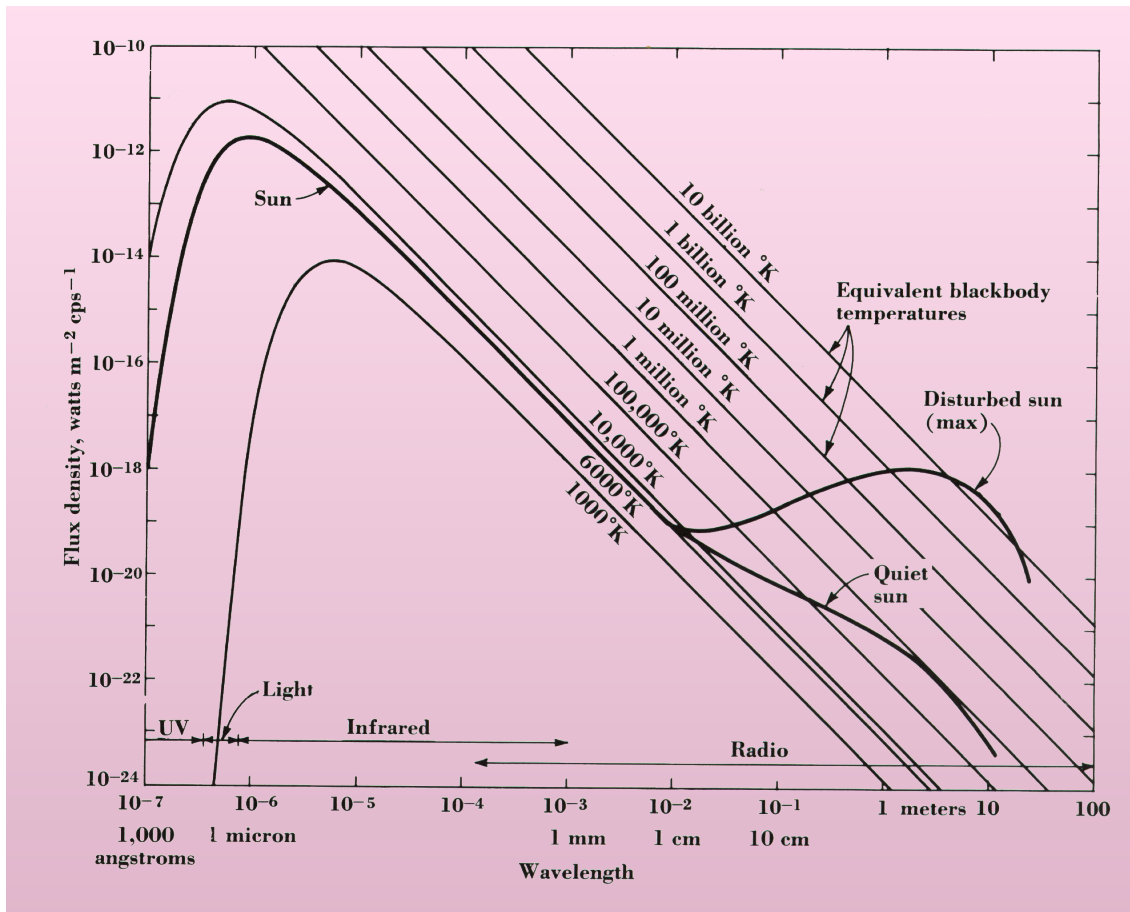


Figure 12: Radiation of the Sun in the radio range compared to the expected thermal radiation of a body of the same size with different temperatures. *Source: Kraus (1986).*

- the slowly variable component: time scale  $t < \text{hours}$ .

This radio radiation is in the ‘quiet’ phase about 100 to 1000 times stronger than the expected thermal radiation of a black body of the temperature of the sun. During the ‘disturbed’ (active) phase the radio flux increases even more. It can reach more than a thousand times the radio flux of the ‘quiet’ Sun (Fig. 12).

The radio flux is so high that a separate unit has been established for it: **The solar flux unit is defined as  $1 \text{ sfu} = 10^{-22} \text{ W m}^{-2} \text{ Hz}^{-1} = 10000 \text{ Jy}$ .**

The radiation of the **quiet sun** is bremsstrahlung of thermal electrons, which are accelerated in the electric fields of ions.

The slowly variable component ( $\lambda 3\text{--}60 \text{ cm}$ ) is associated with condensations ‘radio plagues’ above the photosphere, related to the number and size of sunspots. Strongly circularly polarised, therefore this component is not thermal in origin.

The rapidly variable component shows bursts of radiation lasting from seconds to hours, which are often associated with the occurrence of **flares** in the chromosphere. Flares are sudden in-



creases in light intensity in the vicinity of sunspots. All 2 were at times of high activity. Increased outflow of matter (solar wind) associated with it.

The radio emission from the Sun is constantly monitored by various radio observatories around the world. The *Learmonth Solar Observatory* in Australia publishes a daily radio fluxes of the Sun in 8 frequency bands from 245 to 15400 MHz which can be viewed online<sup>3</sup>. In addition, the Space Weather *Prediction Center (SWPC)* of the US National Weather Service provides on its web page<sup>4</sup> provides current and monthly averaged radiometry data of the Sun since 1991.

Solar storms begin with emission at high frequencies. Over time, as electrons become lower in energy, emission shifts to lower frequencies until it can no longer be detected below 18 MHz. The ongoing monitoring of the solar wind is also called **Space Weather**.

## 6 21-cm radiation and the structure of the Milky Way

### 6.1 History

The ‘father of radio astronomy’ is the US-American engineer Karl G. Jansky, who in 1933 for the first time detected radio waves from space. The discovery was accidental, when Jansky investigated which disturbances are to be expected with transatlantic telephone connections over ‘short wave’ ( $\lambda$  10–20 m).

In the Netherlands, the astronomer Jan Oort (1900–1992) followed Jansky’s work. He realised that spectral lines in the radio range would facilitate the study of the structure of the Milky Way because they would not be blocked by dust within the Galactic plane. Using the Doppler effect, one would be able to measure velocities and map the Milky Way using distance assumptions.

He instructed his student H. C. van der Hulst to find out which suitable spectral lines could be considered. Van der Hulst started with hydrogen, the most abundant element in the universe. He found that there was a suitable transition at 1420.4 MHz, the 21cm Linie. Van der Hulst’s prediction was published in 1945, and a race began to prove this line. It was finally discovered in 1951.

### 6.2 The 21-cm line of neutral hydrogen

In the temperature range from some 10 K up to some 10,000 K, hydrogen is present in atomic form (H I). At even higher temperatures, the hydrogen is ionized (H II). In very cold regions (< few 10 K), e.g. in the centres of molecular clouds, hydrogen is molecular (H<sub>2</sub>).

The transition at the rest frequency  $\nu_0 = 1420.406$  MHz is a ‘hyper-fine structure’ transition in the ground state of atomic hydrogen. The electron can have a magnetic moment either parallel

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<sup>3</sup>[https://www.sws.bom.gov.au/World\\_Data\\_Centre/1/10](https://www.sws.bom.gov.au/World_Data_Centre/1/10)

<sup>4</sup><http://www.swpc.noaa.gov>

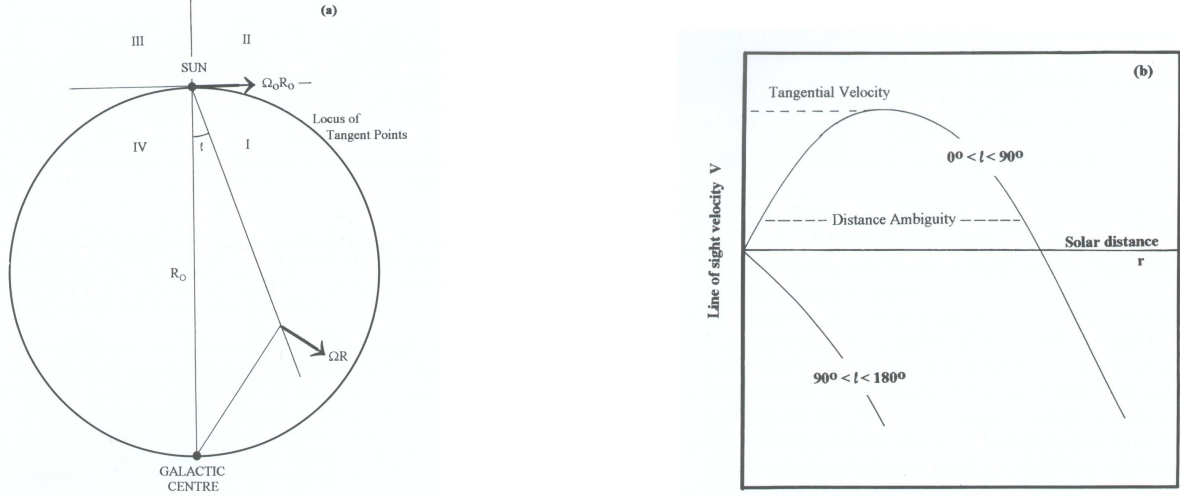


Figure 13: *Left:* Geometry. *Right:* Expected radial velocities in the direction of different galactic longitudes.

or anti-parallel to that of the proton, the parallel state being slightly more energetic than the anti-parallel one. The transition then produces the 21cm line.

The change of direction of the magnetic moment occurs spontaneously only very rarely. The Einstein coefficient is  $A = 2.85 \times 10^{-15} \text{ s}^{-1}$ , i.e. a transition occurs only every 11 million years.

In fact, however, perturbations in the interstellar medium ( $n \approx \text{cm}^{-3}$ ) determine the population of the two levels, so that transitions occur every 300–400 years.

Because of the abundance of hydrogen, one nevertheless receives radiation from all parts of the Milky Way. A spectral line measurements is shifted in frequency by the Doppler effect. When observing the H I line in a certain observation direction, one obtains an overlay for all emitting regions. Due to the different speeds at which the H I clouds move, a sometimes very broad ( $> 100 \text{ km s}^{-1}$ ) line profile is obtained. Complete mapping of the sky has been done in the 70s and 80s (cf. Fig. 14).

If the H I distribution is optically thin at all velocities, the column density of hydrogen can be determined directly from the measured radiation temperature:

$$N_H = 1.823 \times 10^{18} \int T_b dv \quad (87)$$

with

- $N_H$  = Column density in  $\text{cm}^{-2}$
- $T_b$  = Brightness temperature in K
- $dv$  = velocity interval in  $\text{km s}^{-1}$

The condition **optically thin** is generally observed along the Galactic plane  $|b| > 5^\circ$ .

Large optical depths are found along the Galactic plane.

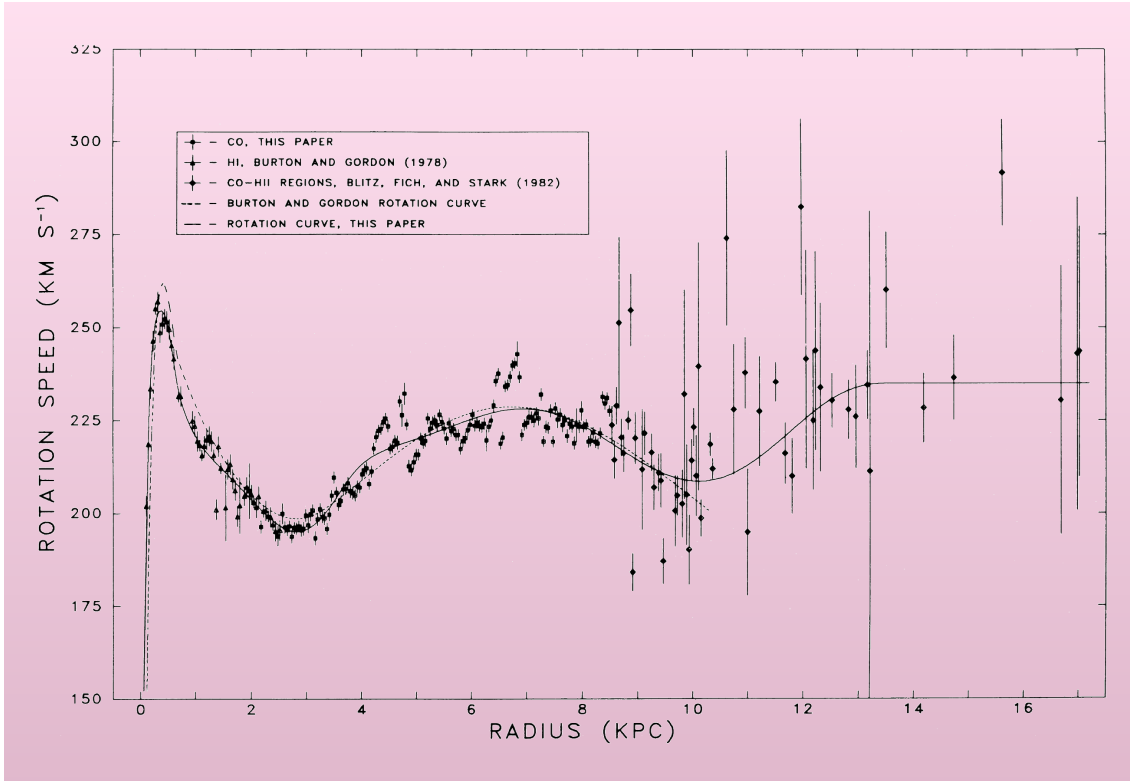


Figure 14: Rotation curve of the Milky Way. It is considered constant between 3 and 8.5 kpc. *Source: Clemens (1985).*

### 6.3 Milky Way rotation curve from H I measurements

The interpretation of the velocities measured by H I can be made under the assumption that the gas runs on circular orbits around the Milky Way centre and that the angular velocity  $\Omega(R)$  decreases with increasing distance from the centre. Then, for the radial velocity  $v_{\text{rad}}$  between a location in the Milky Way and the Sun (cf. Fig. 13):

$$v_{\text{rad}} = R_0 [\omega(R) - \omega_0] \sin(l) \quad (88)$$

with

- $v_{\text{rad}}$  = radial velocity in  $\text{km s}^{-1}$
- $R_0$  = distance Sun–Galactic centre in kpc
- $\omega_0$  = angular velocity at the location of the Sun
- $\omega(R)$  = angular velocity at distance  $R$  from Galactic centre
- $R$  = distance from the Galactic centre
- $l$  = Galactic longitude

Determination of  $\omega(R)$ : For this purpose, the Milky Way is divided into different regions:

- $R < 3 \text{ kpc}$  'Bulge' region
- $3 \text{ kpc} < R < R_0$  Inner Galaxy
- $R > R_0$  Outer Galaxy

The structure in the bulge region is difficult to determine, since non-circular motions of similar magnitude to the circular ones have been found. Possibly the Milky Way has a bar there, whose gravitational potential causes the additional velocities.

The structure of the **inner Galaxy** can be obtained by the following consideration (cf. Fig. 13). If one observes a point on the line of sight along the Galactic length  $l$ , then holds:

- First, the distance  $r$  to the Sun increases and  $R$  decreases, i.e.  $\omega(R)$  increases and so does the radial velocity  $v_{\text{rad}}$ ;
- The maximum speed is reached at the **tangential point**  $R = R_{\text{min}} = R_0 \sin(l)$ ;
- As  $r$  increases,  $\omega(R)$  decreases again and so does  $v_{\text{rad}}$ .

$\omega(R)$  can be determined by assigning the maximum radial velocity (‘terminal velocity’) in the observed profile to the radial velocity at the tangential point.

The terminal velocity method cannot be applied to the outer Galaxy. Instead, one uses  $O$  and  $B$  stars in H II regions, whose distance can be determined with spectro-photometric methods.

It is found that the rotation curve  $v(R)$  between about 3 and 8.5 kpc (Sun) distance from Galactic centre is approximately constant (cf. Fig. 14), because for a point-like mass in the centre  $v(R) \propto 1/\sqrt{R}$  is valid. The conclusion is that the Milky Way, like other galaxies, contains additional non-visible matter. This is the so-called **dark matter**, whose nature is still completely unknown. Since  $v(R) = \text{const.} = \omega(R)R$ ,  $\omega(R)$  decreases with increasing distance from the Galactic centre. This means that the Milky Way has a **differential rotation**.

Current standard values are  $R_0 = 8.5$  kpc, and  $\omega_0 R_0 = 220 \text{ km s}^{-1}$ . From these, the mass of the Milky Way can be estimated:

$$M_{\text{MW}} = 1.1 \times 10^{12} \left( \frac{\omega_0 R_0}{220 \text{ km s}^{-1}} \right)^2 \left( \frac{R_{\text{MW}}}{100 \text{ kpc}} \right) M_{\odot}, \quad (89)$$

in which the contribution of dark matter is also included. Here  $R_{\text{MW}}$  is the radius of the Milky Way including the dark matter halo.

## A Solid angle and angular diameter of a star

Consider a stellar disc of radius  $R$  at a distance  $r$ , where  $R \ll r$  holds. The angular diameter  $\Theta$  of the disc amounts to:

$$\tan \left( \frac{\Theta}{2} \right) = \frac{R}{r}. \quad (90)$$

For small angles,  $\tan(\frac{\Theta}{2}) \approx \frac{\Theta}{2}$  (small angle approximation), so that with the linear diameter  $D = 2R$  follows:

$$\Theta = \frac{D}{r} . \quad (91)$$

The solid angle  $\Omega$ , which this disk spans, is related to the solid angle of the whole sphere  $4\pi$ , as the area of the disk  $\pi (D/2)^2$  is related to the surface area of the sphere with radius  $r$ :

$$\frac{\Omega}{4\pi} = \frac{\pi (D/2)^2}{4\pi r^2} = \frac{\pi \left(\frac{\Theta r}{2}\right)^2}{4\pi r^2} \quad (92)$$

Thus:

$$\Omega = \pi \frac{\Theta^2}{4} , \quad (93)$$

or with  $\Theta = 2R/r$  (cf. equation 91):

$$\Omega = \pi \frac{R^2}{r^2} . \quad (94)$$

## B Recommended textbooks

For further studies, we recommend a range of textbooks that allow the reader a better understanding of radio astronomy. This is, however, not needed in order to perform the F-Praktikum experiment.

- J.D. Kraus, Radio Astronomy, 2nd. Ed., 1986, Cygnus-Quasar Books (good introduction to basics, out of print, see libraries)
- Gerrit L. Verschuur, The Invisible Universe, The Story of Radio Astronomy, 3rd Ed., 2015, Springer, ca. 30 Euro (popular in science)
- Bernard F. Burke, Francis Graham-Smith, An Introduction to Radio Astronomy, 3rd Ed., 2009, Cambridge University Press, ca. 80 Euro (overview, knowledge of physics is assumed, suitable for advanced studies)
- T.L. Wilson, K. Rohlfs, S. Hüttemeister, Tools of Radio Astronomy, 5th Ed., 2008, Springer, available online (within the UHH network) at [SpringerLink](#)
- Wilson T.L. Hüttemeister S., Tools of Radio Astronomy – Problems, 2nd. Ed., 2005, Springer, ca. 45 Euro (collection of problems)
- George Heald, John McKean, Roberto Pizzo, Low Frequency Radio Astronomy and the LOFAR Observatory (especially for LOFAR, for advanced students, available online (within the UHH network) at [SpringerLink](#)